Integration and Kinematics

ANSWER KEY

Section 1: Indefinite Integration Techniques

1 Find the indefinite integral $\int (3x-1)^4\,dx$.

a. $\frac{(3x-1)^5}{15}+C$ ✓

b. $\frac{(3x-1)^5}{5}+C$

c. $\frac{(3x-1)^3}{3}+C$

d. $12(3x-1)^3+C$

Substitute $u=3x-1, du=3dx$: $\int u^4\frac{du}{3}=\frac{u^5}{15}+C=\frac{(3x-1)^5}{15}+C$.

2 Evaluate $\int (e^{2x}+\sin 3x)\,dx$.

a. $\frac{1}{2}e^{2x}-\frac{1}{3}\cos 3x+C$ ✓

b. $\frac{1}{2}e^{2x}+\frac{1}{3}\cos 3x+C$

c. $2e^{2x}-3\cos 3x+C$

d. $e^{2x}-\cos 3x+C$

$\int e^{2x}dx=\frac{1}{2}e^{2x}$, $\int \sin 3x dx=-\frac{1}{3}\cos 3x$.

3 A curve passes through $(1,4)$ and satisfies $\frac{dy}{dx}=4x-1$. Find the equation of the curve.

a. $y=2x^2-x+3$ ✓

b. $y=2x^2-x+4$

c. $y=4x^2-x+1$

d. $y=2x^2+x+1$

$y=\int(4x-1)dx=2x^2-x+C$. At $(1,4)$: $2(1)-1+C=4\Rightarrow C=3$.

Section 2: Definite Integrals and Area Under Curves

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4 Evaluate the exact value of the definite integral $\int_1^4 \left(2x + \frac{1}{\sqrt{x}}\right) dx$.

a. 17 ✓

b. 15

c. 19

d. 16

Antiderivative: $x^2 + 2\sqrt{x}$. At 4: $16 + 2(2) = 20$. At 1: $1 + 2(1) = 3$. Difference: $20 - 3 = 17$.

5 Find the area enclosed by the curve $y = 6x - x^2$ and the x-axis.

a. 36 ✓

b. 18

c. 72

d. 24

Roots: $x(6 - x) = 0 \Rightarrow 0$ and 6. Area $= \int_0^6 (6x - x^2)dx = [3x^2 - x^3/3]_0^6 = 108 - 72 = 36$.

6 Evaluate $\int_0^{\frac{\pi}{4}} \sec^2 x \, dx$.

a. 1 ✓

b. $\sqrt{2}$

c. $\frac{1}{2}$

d. $\pi/4$

$[\tan x]_0^{\pi/4} = \tan(\pi/4) - \tan(0) = 1 - 0 = 1$.

Section 3: Area Between Two Intersecting Curves

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7 Find the area of the finite region enclosed between the parabola $y = x^2$ and the line $y = 2x$.

a. $\frac{4}{3}$ ✓

b. $\frac{2}{3}$

c. 2

d. $\frac{8}{3}$

Intersections: $x^2 = 2x \Rightarrow x = 0, 2$. Area $= \int_0^2 (2x - x^2)dx = [x^2 - x^3/3]_0^2 = 4 - 8/3 = 4/3$.

8 Find the area between the curves $y = 4 - x^2$ and $y = x^2 - 4$.

a. $\frac{64}{3}$ ✓

b. $\frac{32}{3}$

c. 16

d. 8

Intersections: $4 - x^2 = x^2 - 4 \Rightarrow 2x^2 = 8 \Rightarrow x = \pm 2$. Area $= \int_{-2}^{2} (8 - 2x^2)dx = [8x - 2x^3/3]_{-2}^{2} = (16 - 16/3) - (-16 + 16/3) = 64/3$.

9 Set up and evaluate the integral representing the area between $y = \sqrt{x}$ and $y = x$ for $0 \le x \le 1$.

a. $\frac{1}{6}$ ✓

b. $\frac{1}{3}$

c. $\frac{1}{2}$

d. $\frac{2}{3}$

$\int_0^1 (x^{1/2} - x)dx = [\frac{2}{3}x^{3/2} - \frac{x^2}{2}]_0^1 = \frac{2}{3} - \frac{1}{2} = \frac{1}{6}$.

Section 4: Rectilinear Motion and Kinematics

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10 A particle moves in a straight line with velocity $v(t) = 3t^2 - 12t + 9$ m/s. Find the displacement from $t = 0$ to $t = 3$.

a. 0 m ✓

b. 9 m

c. 18 m

d. 6 m

$s(3) - s(0) = \int_0^3 (3t^2 - 12t + 9)dt = [t^3 - 6t^2 + 9t]_0^3 = 27 - 54 + 27 = 0$ m.

11 Find the acceleration of the particle in question 10 at $t = 4$ seconds.

a. 12 m/s^2 ✓

b. 6 m/s^2

c. 24 m/s^2

d. 0 m/s^2

$a(t) = v'(t) = 6t - 12$. At $t = 4$: $a(4) = 24 - 12 = 12$ m/s^2.

12 Calculate the total distance travelled by the particle with $v(t) = 3(t - 1)(t - 3)$ from $t = 0$ to $t = 3$.

a. 8 m ✓

b. 0 m

c. 4 m

d. 12 m

Velocity is positive on $(0, 1)$ ($s(1) = 1 - 6 + 9 = 4$) and negative on $(1, 3)$ ($s(3) - s(1) = 0 - 4 = -4$). Distance $= |4| + |-4| = 8$ m.

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